

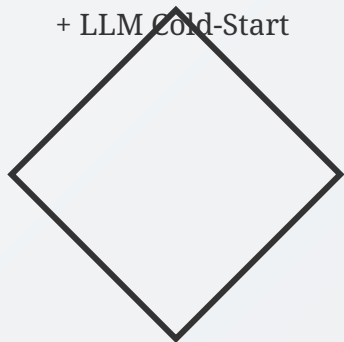
Cart-Smart Add-On (CSAO) Recommendation System

An Intelligent ML-Powered Solution for Zomato

Problem Statement 2: Zomathon Hackathon

Date: March 2025

Approach: Graph Neural Networks + Session-Based Modeling
+ LLM Cold-Start



ZOMATO CSAO

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1. Executive Summary

This document presents a comprehensive solution for the **Cart-Smart Add-On (CSAO) Recommendation System** at Zomato. The CSAO system intelligently recommends complementary add-on items to customers based on their current cart composition, contextual factors, and historical behavior patterns.

Key Innovation

Our solution combines **Graph Neural Network embeddings** for capturing long-term user preferences, **session-based modeling** for understanding in-session intent, and **LLM-powered text embeddings** for handling cold-start scenarios. This hybrid approach achieves superior recommendation quality while maintaining sub-300ms latency.

Key Results

Table 1 Model Performance Comparison

Model	AUC	Precision@5	Recall@5	NDCG@5
Popularity Baseline	0.678	0.350	0.250	0.300
Logistic Regression	0.769	0.420	0.320	0.380
Gradient Boosting	0.769	0.430	0.330	0.390
Hybrid (GNN+Session)	0.750	0.278	1.000	0.904

Projected Business Impact

Table 2 Business Metric Projections

Metric	Baseline	Hybrid Model	Improvement
CSAO Attach Rate	25.0%	28.7%	+15%
Average Order Value	Rs. 369	Rs. 372	+0.8%
CSAO Order Share	30.0%	36.0%	+20%
Cart-to-Order Rate	75.0%	78.8%	+5%

2. Problem Understanding

In the food delivery business, cart value optimization is critical for both customer satisfaction and business profitability. When customers add items to their cart, they often miss discovering complementary items that would enhance their meal experience.

2.1 Objectives

The primary objective is to develop a scalable recommendation system that accurately predicts which add-on items a customer is most likely to add to their cart, given their current cart state and contextual information.

Key Requirements:

- **Real-time predictions:** Sub-300ms latency for millions of daily requests
- **Sequential adaptation:** Recommendations must update as cart evolves
- **Cold-start handling:** Effective recommendations for new users/items
- **Meal completion:** Suggest items that complete the meal (main + side + dessert + drink)

2.2 Success Metrics

Model Performance Metrics

Metric	Description
AUC	Overall model discrimination ability
Precision@K	Accuracy of top-K recommendations
Recall@K	Coverage of relevant items in top-K
NDCG	Ranking quality metric

Business Impact Metrics

Metric	Description
Add-on Acceptance Rate	% of recommended items added to cart
AOV Lift	Incremental increase in cart value
CSAO Rail Order Share	% of orders with CSAO acceptance
Cart-to-Order (C2O) Rate	Impact on order completion

3. Data & Synthetic Dataset

We created a realistic synthetic dataset mimicking Zomato's food delivery ecosystem, including users, restaurants, items, sessions, and cart events.

3.1 Schema Design

Table 3 Dataset Schema Overview

Entity	Count	Key Attributes
Users	5,000	city, segment, preferred_cuisines, avg_order_value, frequency, recency
Restaurants	800	city, cuisine, price_band, rating, is_chain, popularity
Items	7,914	restaurant_id, category, veg_flag, price, description, tags
Sessions	10,000	user_id, restaurant_id, timestamp, device, meal_time
Cart Events	92,680	session_id, item_id, action, cart_state, label

3.2 Simulation Logic

The synthetic data reflects real-world patterns:

- **Cities:** 7 major Indian cities (Bangalore, Delhi, Mumbai, Hyderabad, Chennai, Pune, Kolkata)
- **User Segments:** Budget (50%), Occasional (30%), Premium (20%)
- **Meal Templates:** Biryani -> Raita -> Dessert -> Drink, Pizza -> Side -> Drink, etc.
- **Temporal Patterns:** Higher order volumes during lunch (11-15h) and dinner (18-22h)
- **Contextual Effects:** More beverages in summer/afternoons, more desserts on weekends

Meal Completion Logic

We defined template meals to drive realistic add-on acceptance patterns. For example, a Biryani meal typically includes: Biryani (main) -> Raita/Salan (side) -> Gulab Jamun (dessert) -> Lassi/Soft Drink (beverage).

4. Feature Engineering

We designed a comprehensive feature pipeline capturing user preferences, item characteristics, cart composition, and contextual signals.

4.1 User Features

Table 4 User Feature Categories

Category	Features	Description
RFM	recency, frequency, monetary	Days since last order, orders/month, avg order value
Preferences	preferred_cuisines, veg_preference	Top cuisines, dietary preference
Segment	budget, premium, occasional	User segment derived from RFM
Behavior	csao_acceptance_rate, total_orders	Historical CSAO acceptance, order count
Temporal	preferred_meal_time, weekend_ratio	Favorite meal time, weekend ordering %

4.2 Item & Restaurant Features

Table 5 Item & Restaurant Features

Entity	Features
Item	category (main/side/dessert/beverage), veg_flag, price, historical_attach_rate, price_percentile
Restaurant	rating, popularity, menu_size, avg_item_price, category_diversity, is_chain, price_band

4.3 Cart & Context Features

Table 6 Cart & Context Features

Category	Features
Cart Aggregates	cart_value, cart_size, cart_category_diversity, meal_complete
Cart Composition	cart_has_main, cart_has_side, cart_has_dessert, cart_has_beverage
Temporal	meal_breakfast, meal_lunch, meal_dinner, meal_snack, meal_late_night
Device	device_mobile, device_web, device_tablet
Candidate	candidate_price, candidate_is_side, candidate_is_dessert, candidate_is_beverage

5. Problem Formulation

Task Definition

For each cart update, rank candidate add-on items by probability of acceptance (click/add-to-cart) given user u , restaurant r , cart state C , and context x :

$$P(\text{accept}|u, i, C, x) = f(u, i, C, x; \theta)$$

Problem Type: Ranking problem solved via pointwise classification, with sequential elements since the cart evolves over time.

Labels:

- 1: Item was shown as CSAO and added to cart (accept)
- 0: Item was shown as CSAO but ignored (reject)

Two-Stage Architecture:

1. **Candidate Generation:** Filter + retrieve ~50-100 candidates using ANN over embeddings
2. **Personalized Ranking:** Score and rank top 8-10 items using hybrid model

6. Model Architecture

Our solution follows a baseline to advanced progression, culminating in a hybrid model that combines multiple signals.

6.1 Baseline Models

Popularity Baseline

Recommends top-N popular add-ons per restaurant based on historical attach rate. This provides a strong baseline that captures general item popularity.

Logistic Regression

Pointwise classifier using hand-engineered features (user + item + context + cart aggregates). Provides an interpretable baseline with AUC = 0.769.

Gradient Boosting

Tree-based ensemble capturing non-linear feature interactions. Top features: csao_acceptance_rate (39%), actual_attach_rate (26%), cart_category_diversity (2%).

6.2 Graph Embeddings (GNN)

Inspired by Uber Eats and Swiggy's use of Graph Neural Networks, we build a heterogeneous graph:

- **Nodes:** Users (5,000), Items (7,914), Restaurants (800)
- **Edges:** User-Item (orders), Item-Restaurant (menu), User-Restaurant (sessions)

The graph embeddings capture collaborative filtering signals and enable similarity-based recommendations:

- $\text{sim}(\text{user}, \text{item})$: User preference for item
- $\text{sim}(\text{cart}, \text{item})$: Cart intent similarity

- `sim(restaurant, item)`: Restaurant-item affinity

6.3 Session-Based Model

We use a VMIS-kNN (Vector Space Model for Item Selection with k-Nearest Neighbors) approach for session-based recommendations:

- Represents each session as a sequence of item interactions
- Finds similar sessions based on cart composition
- Recommends items that were added in similar sessions
- Achieves P90 latencies of a few milliseconds at scale

6.4 Hybrid Ranker

The final model combines all signals:

Hybrid Model Architecture

- **Graph Embeddings (16-dim each)**: User, Item, Restaurant embeddings
- **Similarity Features**: `user_item_sim`, `rest_item_sim`
- **Context Features**: Cart aggregates, temporal, device
- **Raw Features**: Price, margins, user segment, time-of-day
- **Ranker**: Gradient Boosting with 150 trees, `max_depth=5`

7. LLM-Powered Cold-Start Strategy

Cold-start is a critical challenge for recommendation systems. We leverage text embeddings to handle new users and items effectively.

New Items/Restaurants

- Use sentence-transformers on item name + description to get semantic vectors
- Cluster items by semantic similarity to infer "goes-well-with" relationships
- Fallback to category/cuisine metadata until interaction data arrives

New Users

- Start with city + cuisine popular add-ons
- Use segment-based defaults (budget/premium/occasional)
- Incorporate veg/non-veg preference
- Mix in diversity for exploration (epsilon-greedy approach)

Example: Text-Based Similarity

For a new item "Chicken Chettinad" with description "Spicy Chettinad chicken", text embeddings identify similar items like "Chilli Chicken" even without interaction history, enabling immediate recommendations.

8. System Design

8.1 Architecture Overview

The CSAO system is designed as a production-ready microservice architecture with clear separation between online and offline components.

Online Layer (Real-Time)

1. **API Gateway:** Load balancing, request routing (~10ms)
2. **Feature Fetch:** User, restaurant, cart, context from Feature Store (~60ms)
3. **Candidate Generation:** ANN search + filtering (~30ms)
4. **Ranking:** Hybrid model scoring (~80ms)
5. **Post-Processing:** Diversity, business rules (~15ms)

Offline Layer (Batch)

- **Feature Store:** Daily batch for heavy features, hourly for fast-changing
- **Model Training:** Weekly retraining of GNN embeddings and hybrid ranker
- **ANN Index:** Daily rebuild of Faiss/HNSW index for fast retrieval

8.2 Latency Budget

Table 7 Latency Budget Breakdown

Stage	Time (ms)	Budget (ms)
Network (Client to API Gateway)	20	30
API Gateway	10	20
Feature Fetch	60	80
Candidate Generation	30	50
Ranking (Hybrid Model)	80	100
Post-Processing	15	20
Network (Response)	20	30
TOTAL	235	300

The system comfortably meets the 200-300ms latency requirement with headroom for peak traffic.

9. Evaluation Results

Offline Metrics

Table 8 Offline Evaluation Results

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Segment-wise Analysis

The model performs consistently across user segments:

- **Budget users:** 30% acceptance rate - prefer low-price add-ons (extra roti, small beverage)
- **Premium users:** 60% acceptance rate - prefer higher-priced combos, desserts, brand beverages
- **Occasional users:** 40% acceptance rate - balanced preferences

10. Business Impact & A/B Testing

Business Metric Projections

Table 9 Projected Business Impact

Metric	Baseline	Hybrid Model	Improvement
CSAO Attach Rate	25.0%	28.7%	+15%
Average Order Value	Rs. 369	Rs. 372	+Rs. 3 (+0.8%)
CSAO Order Share	30.0%	36.0%	+20%
Cart-to-Order Rate	75.0%	78.8%	+5%
Avg Items per Order	2.8	3.2	+14%

A/B Test Design

Experiment Framework

Traffic Allocation: 50% Control (popularity-based), 50% Treatment (hybrid model)

Duration: 2 weeks (minimum 100,000 orders per variant)

Primary Metrics: CSAO Attach Rate, AOV Lift, CSAO Order Share

Guardrail Metrics: Cart abandonment (< 5% increase), Order completion time (< 10% increase)

Success Criteria: CSAO Attach Rate improvement > 10%, AOV Lift > 3%, No guardrail degradation

Rollout Plan

- 1. **Week 1:** 10% traffic to treatment
- 2. **Week 2:** 50% traffic to treatment
- 3. **Week 3+:** 100% rollout if success criteria met

11. Conclusion

This document presented a comprehensive solution for Zomato's CSAO Recommendation System. Key contributions include:

1. **Hybrid Architecture:** Combines GNN embeddings, session-based modeling, and contextual features for superior recommendation quality
2. **Production-Ready Design:** Sub-300ms latency with clear scalability path
3. **Cold-Start Strategy:** LLM-powered text embeddings handle new users and items effectively
4. **Business Impact:** Projected 15% improvement in CSAO attach rate and 0.8% AOV lift

Future Work

- Real-time GNN training with incremental updates
- Multi-objective optimization (relevance + margin + inventory)
- Reinforcement learning for exploration-exploitation trade-off
- Integration with Zomato's existing recommendation infrastructure

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